

Technical Document #130: Software Engineering - Comprehensive Overview

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Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- API design defines how software components communicate.
- Technical debt represents the cost of choosing easy solutions over better ones.
- Microservices architecture structures applications as independently deployable services.